5. The Red Metals

Copper, brass and bronze are known as the 'red metals'. This is because they stand out against other metals due to their characteristic reddish colour.

From construction and architecture to telecommunications and machinery, there is no denying the importance of these 'red metals' in our daily lives.

However, whilst they might all look similar, copper, brass and bronze have very different compositions and properties.

Copper

- Most of the world's copper is extracted from copper ores. Once extracted, copper has many uses in its pure form.
- In addition to its various applications, copper is present in the other red metals – brass and bronze.
- Copper is best known for its electrical and thermal conductivity, malleability and ductility, and resistance to corrosion.
- Today it is most commonly found in electrical materials, roofing, plumbing and industrial machinery.

Brass

- Brass is made by adding varying amounts of zinc to copper.
- Depending on the zinc content, different types of brass can be created and used for different applications.
- The more zinc that is added to the brass, the stronger, more ductile, more malleable and lighter in colour it becomes.
- Brass is commonly used in architecture for its decorative features as well as in manufacturing, construction and in the electrical and plumbing industries.

Bronze

- Bronze is made by mixing copper with tin and other metals such as aluminium, zinc and manganese.
- The properties of bronze vary, depending on its specific composition.
- Bronze is commonly known for its hardness, as well as being highly ductile, brittle and corrosion resistant.
- Bronze is used in architecture as well as for making sculptures, electrical contacts, machine tools and coins.

Examiner only

(a) Tick (✓) to show whether the properties of copper, brass and bronze are fixed or can vary. [2]

	Properties are fixed	Properties can vary
copper		
brass		
bronze		

(b) Tick (✓) the correct statement. [1]

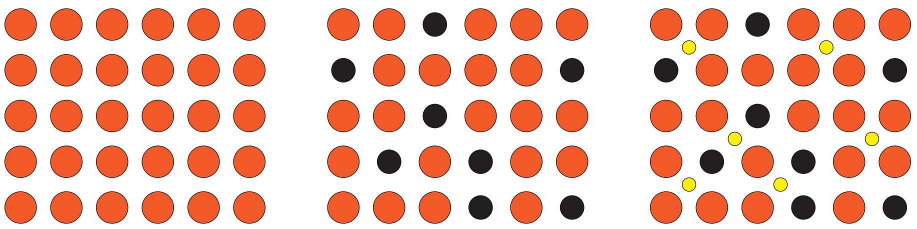
copper, bronze and brass are all metal alloys ☐

copper and bronze are both metal alloys ☐

copper and brass are both metal alloys ☐

bronze and brass are both metal alloys ☐

(c) Give the letter of the structure, **A**, **B** or **C**, that best represents brass. [1]



A **B** **C**

Structure ....

(d) State whether you agree with the following statement. Give a reason for your answer. [1]

'Copper is a good electrical conductor whereas brass and bronze are not'

Agree **Yes** / **No**

Reason ....

....

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